

Lipids, statins and cardiovascular risk:

1. Identifying those who may benefit

(NICE 2023, NG238)



Red Whale

GEMS

Guidelines & Evidence Made Simple

This GEMS does NOT apply to FAMILIAL HYPERCHOLESTEROLAEMIA: see separate article on this.

Suspect if: total cholesterol >7.5 OR personal history CHD <60y OR CHD <60y in first-degree relative.

We are also asked to systematically search our records for those with cholesterol >9 if ≥30y (>7.5 if <30y); very likely to have familial hypercholesterolaemia. *Have you done this yet?* (NICE 2019, CG71)

IDENTIFYING THOSE AT INCREASED RISK OF CVD

How do we assess risk in the person in front of us? Do we use QRISK, or have they got other factors that mean they are high risk by definition? So, the question is: *to QRISK or not to QRISK?*

Secondary prevention of CVD UNLESS have CKD

Do NOT use risk calculator: high risk by definition

Secondary prevention = those with CVD

- **OFFER STATIN: atorvastatin 80mg (regardless of baseline cholesterol)**
- **CONSIDER adding ezetimibe to statin to further reduce risk, even if cholesterol target met**
- Use lower dose if patient preference or if:
 - Potential drug interactions
 - High risk of adverse events

Atorvastatin
80mg

+/-

Ezetimibe

Start statin at diagnosis: do not wait to assess impact of lifestyle changes

CKD (in primary and secondary prevention of CVD) (eGFR <60 or ACR ≥3 on 2 occasions, 3m apart)

Do NOT use
QRISK3:
high risk by
definition

- **OFFER statin to ALL with CKD: for both those WITH and WITHOUT CVD**
- Use atorvastatin 20mg initially for ALL (primary and secondary prevention)
 - eGFR ≥30: if cholesterol target not met, increase dose (primary and secondary prevention)
 - eGFR <30: agree use of higher doses of statin with renal specialist

Atorvastatin
20mg initially

WHY not 80mg for secondary prevention? Limited evidence for statins in CKD. Guidance based mainly on consensus. Lower doses used because of concern about adverse events/licence

Primary prevention of CVD in 3 special groups

Primary prevention = those without CVD

Type 1 diabetes

- Do NOT do risk calculation: high risk by definition
- **OFFER atorvastatin 20mg if any of:**
 - >40y
 - Had diabetes for >10y
 - Other CV risk factors or nephropathy (not defined by NICE)
- **CONSIDER statins for those aged 18–40y with T1DM**

Atorvastatin
20mg

≥85y

- In older people, do NOT do risk calculation: age alone puts them at increased risk, particularly if they smoke or have raised BP. Take frailty into account
- **CONSIDER atorvastatin 20mg**

Atorvastatin
20mg

Living with HIV + ≥40y

- QRISK3 underestimates CVD risk in those living with HIV
- BHIVA tells us to offer everyone ≥40y living with HIV a statin (BHIVA 2024)
- Caution! Antiretroviral therapy can interact with statins (see article *HIV: managing HIV in primary care* for more details)

Atorvastatin
20mg

Primary prevention in everyone else: 25–84y, including T2DM and hypertension, but not those with CVD, CKD, T1DM or ≥85y

Use
QRISK3
to assess
risk

- **If QRISK3 ≥10%: offer atorvastatin 20mg**
- **If QRISK3 <10%: do not rule out statin.** May be used if patient chooses to, or if concern risk is underestimated (on HIV treatment, recently stopped smoking, on drugs to reduce CV risk or drugs that cause dyslipidaemia (e.g. immunosuppressants)). **Until QRISK3 integrated into clinical systems, QRISK2 can be used, BUT use QRISK3 online if any of the following** (QRISK2 will underestimate risk):
 - Severe mental illness
 - On atypical antipsychotics
 - SLE
 - Erectile dysfunction
 - On corticosteroids
 - Migraine

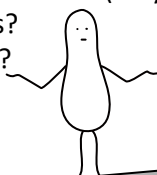
Atorvastatin
20mg

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COMMUNICATING RISK

NICE calculates that statins are cost-effective even if QRISK <10% (the cost of statins to whole groups of people results in cost savings from fewer CV events). This needs balancing against individuals' willingness to take a daily tablet (especially if at low risk). We need to communicate risk carefully to find the right decision for the person before us. If treatment is declined, this can be reassessed again in future. NICE says discussion should include:

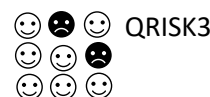
- What is their level of risk?
- What do they already know?
- Health beliefs about what determines future health (may affect attitude to health)?
- Presence/absence of modifiable risk factors?
- Willingness/ability to make lifestyle change?
- Views on medication?



Absolute level of risk varies from person to person
 Individual attitude to risk will vary

When discussing risk, NICE recommends:

- Tell people about their **10y absolute CVD risk** (i.e. QRISK3 percentage). This is expressed in terms of smiley faces on the QRISK3 website
- Consider using QRISK3-lifetime as part of the discussion, especially if QRISK3 <10% or in those <40y with risk factors. QRISK3-lifetime allows you to enter some 'What if..?' data and see the impact such changes have on lifetime risk of CVD, e.g. what if I gave up smoking/lowered my cholesterol/reduced my BP/lost weight?



When to reassess risk? NICE says to review risk on an 'ongoing' basis if >40y. Unhelpful!

In reality, this may depend on magnitude of risk, patient choice and primary care capacity!

LIFESTYLE ADVICE

Remember to offer support with lifestyle change: exercise referral schemes, weight management services, etc.

Smoking Stop smoking.

- Diet** **Cardioprotective diet:** reduce fat intake to $\leq 30\%$ of total energy intake. Swap saturated fats to mono- or polyunsaturated fats (use olive/rapeseed oil). Keep saturated fats to $\leq 7\%$ of total energy intake.
Healthy eating as per NHS Eat Well guide (i.e. reduce sugar intake; use whole grains; low salt (<6g/1tsp/d); ≥ 5 portions/d of fruit and veg; 4 portions/w of unsalted nuts, seeds and legumes; and 2 portions/w of fish (1 of which should be oily fish, i.e. tuna, salmon, trout, mackerel, sardines, pilchards, kipper, eel, whitebait, herring. NOT tinned tuna as all the oil is removed).
Alcohol: ≤ 14 U/w, spreading drinking evenly over 3 or more days. Do not binge.
Do not recommend plant stanols/sterols if being treated for high cholesterol or if CKD, T1DM or T2DM (NICE doesn't say why, but there is a lack of evidence of effectiveness).

Work towards a healthy weight **If overweight and at high risk of CVD, offer interventions in line with NICE obesity guideline (NICE 2023, CG189)** (including medication or bariatric surgery if criteria met; remember, criteria are lower for most non-white ethnicities).

Physical activity 150min/w moderate intensity (light sweat) **or** 75min/w vigorous intensity (hard breathing) **PLUS** muscle strengthening and balance activities 2x a week.
Agree goals and consider exercise referral scheme.

Do NOT routinely offer aspirin for the primary prevention of CVD because the benefits and harms are finely balanced (NICE made this decision having reviewed the evidence from the ARRIVE, ASCEND and ASPREE trials).

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Lipids, statins and cardiovascular risk:

3. Statin initiation, monitoring, targets

(NICE 2023, NG238 and NICE AAC group, 2023)

BEFORE STARTING A STATIN

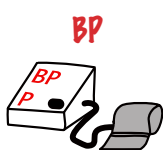
PRIMARY PREVENTION: encourage lifestyle modification, review and consider reassessment of CV risk before starting statins. If lifestyle change is ineffective/inappropriate, offer statin therapy.

SECONDARY PREVENTION: don't delay statin while improving modifiable risk factors: do the two in parallel.

- In primary and secondary prevention: assess for comorbidities and secondary causes of dyslipidaemia:

Lifestyle factors increasing CV risk

Look for obesity (BMI)
Smoking
Alcohol



Bloods (fasting not needed)

Full lipid profile → Total cholesterol
ALT or AST → HDL and non-HDL
HbA1c for diabetes → LDL
Renal function → Triglycerides
TSH if symptoms of hypo/hyperthyroidism

- If patient has ever had persistent, generalised, unexplained muscle symptoms: measure CK
 - If CK >5x upper limit of normal, remeasure in 1w. If still >5x upper limit of normal, do not start statin.
 - If CK raised but <5x upper limit of normal, start statin at a lower dose.
- Advise patients to report muscle pain, tenderness or weakness. If present, check CK: see section 5 of GEMS

Statins are teratogenic: stop 3m before conception, avoid in pregnancy, restart when breastfeeding ceases.

Interactions with other medication: advise that some food (e.g. grapefruit) and some supplements interfere with statins: read patient leaflet and consult the prescriber/pharmacist when starting new drugs/supplements. If statin stopped for treatment for an intercurrent illness (e.g. antibiotics), remember to restart again.

MONITORING STATINS

AT 3 MONTHS (or after changing treatment)

Recheck bloods after 2–3m:

- Full lipid profile
- Transaminases (AST or ALT)



ANNUAL REVIEW

- Review lifestyle and CV risk factors
 - AST or ALT: at FIRST annual review – NOT thereafter
 - Full lipid profile
 - Do if secondary prevention
 - Consider if primary prevention
 - Do NOT measure CK if ASYMPTOMATIC
 - If not on high-intensity statin (<20mg atorva/equivalent), discuss benefits of swapping to high-intensity statin
- If AST/ALT abnormal
- If abnormal but <3x upper limit of normal:
- Continue statin, recheck in 4–6w
- If abnormal and ≥3x upper limit of normal:
- Stop statin, recheck in 4–6w
 - Once normalised, cautiously reintroduce statin (lower dose, different statin)
 - If AST/ALT remain elevated, consider other causes (e.g. alcohol, NAFLD)
- (based on European guidance, EHJ 2020: 41;111)

TARGET ON TREATMENT

PRIMARY PREVENTION

Aim for >40% reduction in non-HDL



SECONDARY PREVENTION

Aim for LDL ≤2mmol/l or non-HDL ≤2.6mmol/l

If target not achieved (primary or secondary prevention):

- Discuss compliance, timing of dose (take at night) and lifestyle modification
- If not on a high-intensity statin at maximum tolerated dose, consider increasing statin dose or swapping to higher-intensity statin. Take into account patient's views, comorbidities, polypharmacy, frailty, life expectancy

And in SECONDARY PREVENTION

- If target not met, consider adding additional drugs (see section 4 of this GEMS)
- Even if target IS met, ezetimibe can be added to statins to reduce CV risk further

Statin intensity and likely LDL reductions

High intensity (LDL fall usually >40%)	Medium intensity (LDL fall usually 30–40%)	Low intensity (LDL fall usually 20–30%)
Atorvastatin ≥20mg	Atorvastatin 10mg	
Rosuvastatin ≥10mg	Rosuvastatin 5mg	
Simvastatin 80mg*	Simvastatin 20–40mg	Simvastatin 10mg
	Fluvastatin 80mg	Fluvastatin 20–40mg
		Pravastatin 10–40mg

* Simvastatin 80mg has higher risk of myopathy than other statins with similar LDL-lowering efficacy. Use is therefore restricted – see main article for details.

From BNF Dyslipidaemias section:
<https://bnf.nice.org.uk/treatment-summaries/dyslipidaemias/>

OTHER LIPID-LOWERING TREATMENTS

Do NOT routinely use/recommend (because of lack of evidence): fibrates, bile acid sequestrants, nicotinic acid (niacin), omega-3 (except icosapent ethyl). Do NOT advise co-enzyme Q₁₀ or vitamin D for statin side-effects.

There are 4 situations in which NICE suggests non-statin lipid-lowering drugs may be used:

Statins contraindicated or genuine intolerance

Ensure genuine intolerance! ‘De-challenge’, ‘re-challenge’ (section 5) for several statins (NICE NG238 and NICE 2016/21 TA385, 694)

Try **Ezetimibe monotherapy**

If target still not met, use **Ezetimibe with bempedoic acid** or (only if CVD) consider **PCSK-9 drug** — If NICE criteria (below) met

Statins not achieving targets

Add **Ezetimibe** to statin*

If CVD (not without), consider **PCSK-9 drug** — if NICE criteria met

*Does this apply to primary or secondary prevention? NICE guidance suggests this is only for SECONDARY prevention, but the NICE ezetimibe TA implies it can be used in primary prevention too

In CVD to reduce risk further (even if cholesterol target met)

In those with CVD, on statins, even if target cholesterol met, **Ezetimibe** can be added to further reduce CV risk

CVD with raised triglycerides

Icosapent ethyl (a form of omega 3) can be considered if (and this is complicated!): **Known CVD AND at high risk of further events AND**

On statins AND Well-controlled LDL (>1.04 but ≤2.6mmol/l) AND Raised fasting triglycerides (≥1.7mmol/l) (NICE 2022, TA805)

When can ezetimibe be used?

Ezetimibe

Putting the information above in a different way, when can ezetimibe be used?

- If statin contraindicated or (genuine!) statin intolerance
- If statin at maximum tolerated dose not achieving cholesterol target
- In secondary prevention, even if target cholesterol met, to reduce CV risk further

When can PCSK9 inhibitors be used?

NICE criteria for PCSK9 inhibitor use: ONLY if CVD AND... (NICE TA393, 394, 733)

Inclisiran	LDL persistently ≥2.6mmol/l
Evolocumab or alirocumab	LDL persistently >4mmol/l or LDL persistently >3.5mmol/l if: recurrent CV events or events in more than one vascular bed (e.g. CVA and CHD)

ABNORMAL LIPID PROFILES: management

If significant dyslipidaemia, consider causes: alcohol, uncontrolled diabetes, hypothyroidism, liver disease, nephrotic syndrome.

Total cholesterol >7.5mmol/l		Consider familial hypercholesterolaemia (FHc), especially if personal/family history (first-degree relative) of CHD <60y: see details in our FHc article
Total cholesterol >9 or non-HDL >7.5		Refer, even in the absence of a family history of premature cardiac disease
Triglycerides	4.5–9.9mmol/l	Optimise management of other risk factors Refer if non-HDL cholesterol >7.5 (CV risk tools will underestimate risk)
	10–20mmol/l	Repeat 5–14d later, after fasting : 30% will be normal on fasting If still raised, consider other causes and seek specialist advice
	>20mmol/l	Refer urgently (if not due to excess alcohol or uncontrolled diabetes)

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5. Adverse events and statin intolerance pathway

(NICE 2023, NG238 and NICE AAC group, 2023)



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ADVERSE EVENTS

Abnormal LFTs on statins. Measure AST/ALT at 3m and 12m, THEN DO NOT ROUTINELY MEASURE AGAIN
Do not routinely withhold statins if raised ALT/AST but <3x upper limit of normal

Muscle symptoms. NICE describes these groups:

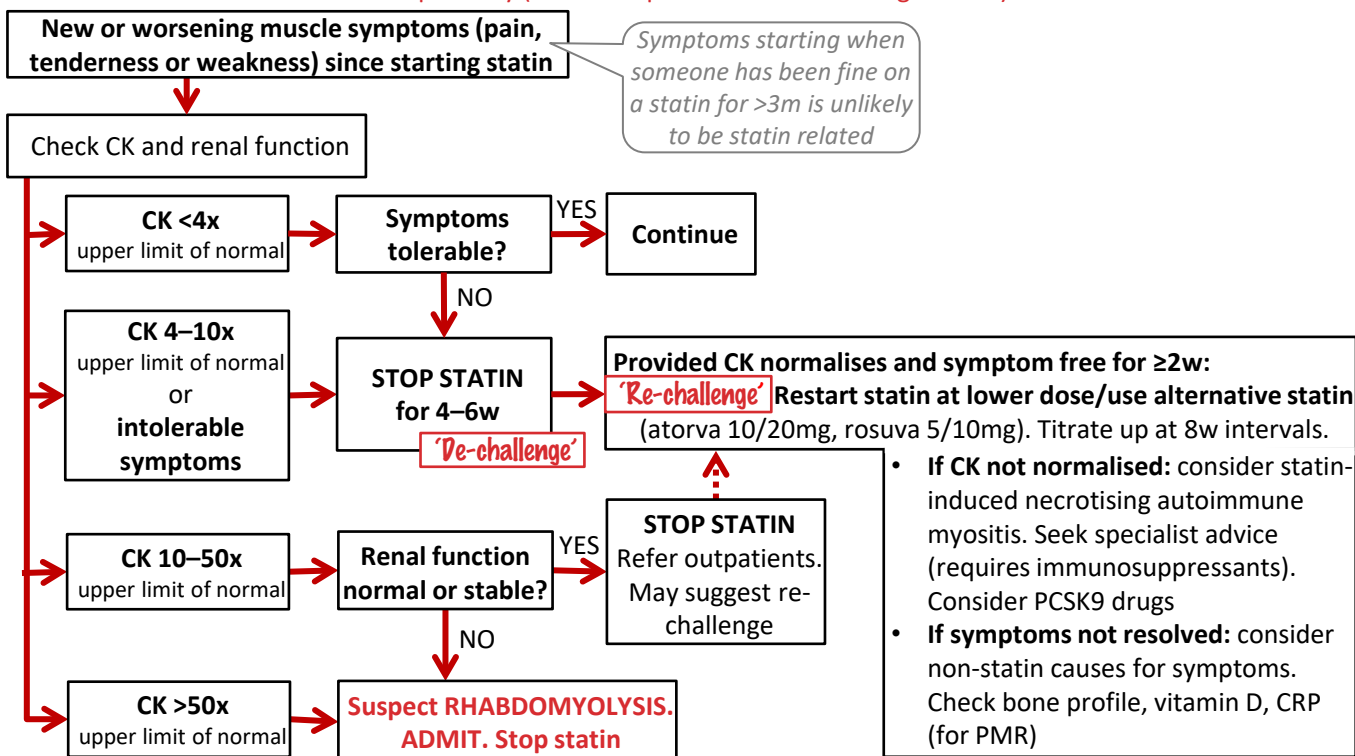
- **Statin reluctance:** hesitation to even start statins because of perceived risks of muscle symptoms
- **Statin-related muscle toxicity:** symptoms come with challenge (starting statin), resolve with **'de-challenge'** (stopping statin) and recur with **'re-challenge'** (restarting). (Ranges from muscle symptoms with normal CK/slightly raised CK with no symptoms, all the way through to rhabdomyolysis and statin-induced necrotising autoimmune myositis.)
- **Statin-associated muscle symptoms:** may not be due to statin-related muscle toxicity: the response to 'de-challenge/re-challenge' seen in statin-related muscle toxicity is NOT seen here
- **Non-statin-related symptoms:** another cause is likely

MANAGEMENT OF MUSCLE SYMPTOMS

Statin have the best evidence for CVD risk reduction. Avoid labelling people as statin intolerant too soon!

- Use a **'de-challenge'** then **'re-challenge'** approach (stop statin, then restart when symptoms settled), and do this with several options **before** labelling person as statin intolerant.

This is summarised from NICE AAC pathway (more comprehensive than NICE guidance)



Aim for maximum tolerated dose (but even low dose will reduce CVD risk)

- If not tolerating daily dose: try alternate-day dosing or twice-weekly dosing using atorvastatin or rosuvastatin (they have longer half-lives)
- Consider using lower-dose statin with ezetimibe
- **If recurrence of intolerable symptoms:** consider other options (ezetimibe, ezetimibe with bempedoic acid, PCSK9 inhibitors/inclisiran (if meets NICE criteria))
- NICE guideline also advises: if in high-risk group (e.g. CVD, T1 and T2 diabetes, CKD, dyslipidaemias) and unable to tolerate 3 different statins, seek specialist advice

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